Habituation as Model Update in a Minimal Embodied Agent: Controls, Dishabituation Limits, and Weight Persistence Without Explicit Memory

Adrián Valerio Porras[1]

1 Independent Researcher, San José, Costa Rica — ORCID: 0009-0003-8445-2214 — adrian@example.com

Conciencia Artificial v0.12 — 30 August 2026 — Preprint. Intended submission to arXiv cs.AI and q-bio.NC

Abstract — We built a tiny embodied agent that lives on a laptop. No vision, no hearing, only a body. It predicts its next bodily state from its current state and its own action, monitors homeostatic variables, and maintains a second model that predicts when its own predictions will fail. We pre-registered a 30-seed study to test whether this minimal loop can detect violations of its sensorimotor contingencies, habituate by updating its model, retain that update in weights after erasing episodic memory, and act on its own uncertainty. It can. Detection was reliable ($z=20.6$, 95% CI [16.0, 25.5]). Habituation cut surprise by 86% ($d=3.5$). The trace survived erasure of explicit memory (post/pre ratio 0.02) and lived in the weight delta. A meta-cognitive module Phi was calibrated ($r=0.701$), generalized out of distribution ($r_{}=0.730$), and causally changed behaviour: the agent left an unreliable zone faster when Phi was coupled to action (15.0% vs 28.1% time, $d=-1.61$). We report one clear failure and share all code. If you want to try it, it runs on a laptop in minutes.

NOTE: This is an auto-generated placeholder PDF for Zenodo drag-and-drop. For the final camera-ready PDF, compile `Habituation_Model_Update_v0.12.tex` on Overleaf (pdflatex ×3) and replace this file before publishing. The TeX source in this folder is the authoritative version (paper/main.tex:1, braces 321/321). All statistics are from `framework/*.py` and `results/*.json` — 30-seed pre-registered, single MacBook no GPU.

Keywords: embodied cognition, habituation, metacognition, homeostatic reinforcement learning, sensorimotor contingencies, model update

Full paper: 10 sections, 21 references, 3 figure placeholders (see TeX). Reproducibility: `python3 framework/organismo_final.py --steps 30000` — see README.md in this folder.

Zenodo: Drag this entire folder's contents (PDF + TeX + LICENSE + CITATION.cff) into the 'Drag and drop files' area above. See ../REPORT_v0.12_FINAL.md:6 for 5-step checklist.